

Home Assignment 3: Comparative Study of Unsupervised Learning Algorithms

Dataset: Wholesale Customers – UCI ML Repository

[Dataset Link – UCI Wholesale Customers](#)

Algorithms to be studied:

1. PCA – Principal Component Analysis
2. K-Means Clustering
3. Hierarchical Agglomerative Clustering
4. GMM – Gaussian Mixture Model
5. DBSCAN

Objective

Use the Wholesale Customers dataset to investigate how different unsupervised learning algorithms discover hidden patterns in customer spending behaviour, and understand why different algorithms produce different results.

Part 1 – Data Understanding & Preprocessing

Tasks

1. Load the dataset.
2. Display:
 - Number of observations
 - Number of features
 - Data types
 - Missing values
 - Basic statistics
3. Separate:
 - Numerical spending features
 - Categorical features
4. Use the six spending attributes for unsupervised learning:
 - Fresh
 - Milk
 - Grocery

- Frozen
 - Detergents_Paper
 - Delicassen
5. Standardize the numerical features.

Students should observe

- Q1.** Which spending attributes have the largest variation?
- Q2.** Are all six features on a comparable scale?
- Q3.** What could happen if clustering is performed **without standardization**?
- Q4.** Which product categories appear to be correlated?
- Q5.** Are there any unusually high-spending customers?

Part 2 – PCA

Apply PCA to the standardized data.

- Calculate all principal components.
- Plot **explained variance ratio**.
- Plot **cumulative explained variance**.
- Determine the number of components required to retain approximately **90% or 95% variance**.
- Create a 2D scatter plot using PC1 and PC2.
- Examine PCA loadings.

Observation Questions

- Q6.** How many principal components are required to represent most of the information?
- Q7.** How much variance is explained by PC1?
- Q8.** Which original features contribute most strongly to PC1?
- Q9.** Which features contribute most strongly to PC2?
- Q10.** Can six-dimensional customer behaviour be reasonably represented using only two principal components?
- Q11.** What information is lost when reducing six dimensions to two?

Important conceptual question

Why is PCA useful before clustering even though PCA itself does not create clusters?

Part 3 – K-Means Clustering

Tasks

Apply K-Means for: $K = 2, 3, 4, 5, 6$

For each K , calculate:

- Inertia
- Silhouette Score

Plot:

- Elbow curve
- Silhouette score vs K

Observation Questions

Q12. At what value of K does the elbow appear?

Q13. Does the Silhouette Score suggest the same K as the Elbow Method?

Q14. If the two methods suggest different K values, which one would you prefer and why?

Q15. Are the K-Means clusters clearly separated in the PCA plot?

Q16. Which cluster contains customers with the highest overall spending?

Q17. Which product categories distinguish the clusters?

Important question

Does K-Means always produce the "correct" number of clusters? Explain using your results.

Part 4 – Hierarchical Clustering

Apply **Agglomerative Hierarchical Clustering**.

Students should experiment with:

- Single linkage
- Complete linkage
- Average linkage
- Ward linkage

Tasks

1. Generate dendrograms.
2. Identify possible cluster counts.
3. Create clusters using the selected number of clusters.
4. Compare the result with K-Means.

Observation Questions

Q18. What does the dendrogram tell you about the similarity between customers?

Q19. At what level would you cut the dendrogram to obtain meaningful clusters?

Q20. Does the dendrogram suggest the same number of clusters as K-Means?

Q21. How does changing the linkage method affect the clusters?

Q22. Which linkage method gives the most meaningful segmentation?

Key conceptual question

Why might Hierarchical Clustering be preferred when we do not know the number of clusters in advance?

Part 5 – Gaussian Mixture Model (GMM)

Apply GMM with: `n_components = 2, 3, 4, 5, 6`

Calculate:

- AIC

- BIC
- Silhouette Score

Also plot the clusters using PCA.

Observation Questions

Q23. Which number of Gaussian components gives the lowest BIC?

Q24. Is the optimal number of components according to BIC the same as the K-Means result?

Q25. Examine the probability assigned to each customer. Do some customers have similar probabilities for two clusters?

Q26. What does this tell you about customers lying near cluster boundaries?

Important conceptual question

K-Means gives a hard cluster assignment, whereas GMM gives a probability of belonging to each cluster. What advantage does this provide in customer segmentation?

Which customer is more ambiguous? Why?

Part 6 – DBSCAN

Tasks

Experiment with different:

- `eps`
- `min_samples`

For example:

`eps = 0.3, 0.5, 0.7, 1.0`

`min_samples = 5, 10`

Students should identify:

- Number of clusters
- Number of noise/outlier points
- Silhouette Score where appropriate

- PCA visualization

Observation Questions

Q27. How many clusters does DBSCAN identify?

Q28. How many observations are classified as noise?

Q29. What happens when eps is increased?

Q30. What happens when eps is decreased?

Q31. Why does DBSCAN not require us to specify the number of clusters beforehand?

Q32. Are the customers identified as noise necessarily "bad data"?

Q33. Could a high-spending customer legitimately be considered an outlier?

Important conceptual question

Why might DBSCAN be more useful than K-Means when the dataset contains outliers or irregularly shaped clusters?

Part 7 – Compare All Algorithms

Algorithm	Number of Clusters	Cluster Quality	Handles Noise?	Hard/Soft	Main Advantage	Main Limitation
K-Means				Hard		
Hierarchical				Hard		
GMM				Soft		
DBSCAN				Hard		

Final Observation Questions

- Q34.** Did all algorithms produce the same number of clusters?
- Q35.** Why do different algorithms produce different cluster structures on the same dataset?
- Q36.** Which algorithm produced the most interpretable customer segments?
- Q37.** Which algorithm was most sensitive to parameter selection?
- Q38.** Which algorithm handled outliers most effectively?
- Q39.** Which algorithm would you choose for this dataset if you were working for the wholesale company?
- Q40.** Justify your choice using **at least three observations from your experiments**.